

ATHARV TEMBHURNIKAR

College Park, MD, USA

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Experience

Teaching Assistant (Agentic AI) – University of Maryland (UMD) **Aug 2026 – Present**

Machine Learning Research Assistant – University of Maryland (UMD) **Dec 2025 – Jul 2026**

- Developed ML-based pipelines for cell-wise annotation and intensity tracking on artificial-nose sensor data, enabling structured capture of cellular responses and temporal patterns over time.
- Engineered interpretable feature representations (response amplitude, decay dynamics, temporal slopes) to quantify cell behavior, supporting data-driven validation of experimental hypotheses and robust pattern classification.

Data Science Intern – Hackveda Private Limited (Pune, India) **Jan 2025 – Jun 2025**

- Analyzed real-time e-commerce transaction data to identify trends in orders, pricing, product categories, and regional demand, supporting improved inventory planning and delivery efficiency.
- Built interactive Power BI dashboards to track sales performance and regional demand, enabling faster monitoring and data-driven operational decisions.
- Performed statistical analysis on pricing and discount strategies to evaluate their impact on order volume and revenue, contributing to optimized promotions and improved inventory turnover.

Research Assistant – University of Pune **Jan 2023 – Dec 2023**

- Contributed to a real-time posture recognition and correction system using CNN and Mediapipe; published findings at [IEEE TEECCON 2024](#).
- Enhanced model performance by over 6% through architectural modifications, improving reliability across diverse demographics and video inputs.

Education

University of Maryland, College Park **Aug 2025 – May 2027**

Master of Science in Data Science *GPA- 4*

University of Pune **Aug 2021 – May 2025**

Bachelor of Technology in Artificial Intelligence and Data Science *GPA- 3.83*

Technical Skills

Programming and Data Science Tools: Python, Java, MySQL, PostgreSQL, MongoDB, Amazon Web Services, NumPy, Pandas, Scikit-learn, PyTorch, Langchain, Langraph, Power BI, Tableau, Git

Domains: Machine Learning, Deep Learning, Natural Language Processing, Gen AI, Agentic AI, Business Intelligence

Other Skills & Methodologies: Research, Mathematics and Statistics, Data Analytics, Agile Methodology

Projects and Research Publications

DCFare | *Python, XGBoost, Neural Networks* **Aug 2025 - Dec 2025**

- * Built an end-to-end predictive pipeline on 2.5M+ DC taxicab trips; performed systematic feature engineering and model selection, comparing ensemble and neural approaches to quantify performance-data scale trade-offs.
- * Achieved R^2 of 0.96 with an MLP neural network; conducted ethics-aware analysis addressing privacy, fairness, and responsible use of mobility data.

EcoIntel | *Python, XGBoost, CNN+LSTM, Gemini API, Streamlit* **Jul 2024 - Dec 2024**

- * Designed a two-stage review analysis pipeline filtering fraudulent and low-quality reviews prior to sentiment analysis, improving sentiment prediction reliability by 11%.
- * Integrated Gemini API for concise review summarization; published at [IEEE ICC 2025](#), demonstrating real-world impact of hybrid deep learning in e-commerce analytics.

SkillSync | *Python, spaCy, NLTK, KNN* **Jan 2024 - May 2024**

- * Built an NLP-driven resume parsing system using spaCy + custom NER pipelines to extract skills, experience, and job-relevant entities; developed an ML classifier improving matching accuracy by 10%+ over baseline.
- * Designed a job-description similarity engine generating percentage-based fit scores, enabling recruiters to quickly assess candidate relevance.

Extracurricular Activities

- * Authored 35+ tech articles on Medium, engaging over 3000+ readers.
- * Finalist at Viz-a-thon organized by Artificial Intelligence Student's Association, VIIT, Pune.
- * Won AI Artist INNOVATION24 Competition organized by Cummins College of Engineering, Pune.